

Smart Attendance System Using Face Recognition

1. Introduction

Attendance tracking is a fundamental task in educational institutions. Conventional methods such as manual registers or RFID-based systems are inefficient and susceptible to proxy attendance. With advancements in Artificial Intelligence and Computer Vision, automated attendance systems have emerged as reliable alternatives.

This project proposes a Smart Attendance System that uses deep learning models for real-time face detection and recognition. It aligns with modern Digital Education Ecosystems by integrating AI-driven automation into classroom management.

2. Problem Statement

Existing attendance systems suffer from:

- Manual errors and inefficiency
- Time consumption in large classrooms
- Lack of real-time monitoring
- Possibility of proxy attendance

The need is for a **secure, automated, and scalable solution** using AI.

3. Proposed System

The proposed system leverages:

- **YOLOv8** → Real-time person detection
- **FaceNet** → Face embedding generation
- **Cosine Similarity Matching** → Identity verification
- **MySQL Database** → Attendance storage
- **Streamlit Dashboard** → Visualization and analytics
- **Email Notification Module** → Automated attendance alerts

Workflow

1. Capture live video from camera
2. Detect persons using YOLO
3. Extract face region
4. Generate embeddings using FaceNet

5. Match with stored embeddings
6. Mark attendance in database
7. Trigger email notification to user/admin
8. Display analytics on dashboard

4. System Architecture

The system consists of:

- **Input Layer:** Camera feed
- **Processing Layer:** YOLO + FaceNet
- **Database Layer:** MySQL
- **Application Layer:** Streamlit UI

5. Implementation Details

- Programming Language: Python
- Frameworks: Streamlit
- Libraries: OpenCV, Ultralytics YOLO, TensorFlow/PyTorch
- Database: MySQL
- Email Services: SMTP (Python-based email alerts)
- Deployment Ready for: Web-based Educational ERP systems

Code Implementation:

```
import streamlit as st
import cv2
import pickle
import pandas as pd
import mysql.connector
from ultralytics import YOLO
from collections import Counter
from utils.face_utils import preprocess_face
from utils.match_utils import match_face
from models.facenet_model import get_embedding
```

```
from attendance.mark_attendance import mark

import sys

import os

sys.path.append(os.path.abspath(os.path.join(os.path.dirname(__file__), '..')))

# LOAD ENV

from dotenv import load_dotenv

load_dotenv()

# DB CONNECTION

conn = mysql.connector.connect(
    host=os.getenv("DB_HOST"),
    user=os.getenv("DB_USER"),
    password=os.getenv("DB_PASS"),
    database=os.getenv("DB_NAME"),
    auth_plugin='mysql_native_password'
)

# LOAD MODELS

yolo = YOLO("yolov8n.pt")

with open("embeddings/embeddings.pkl", "rb") as f:
    database = pickle.load(f)

# UI SETUP

st.set_page_config(page_title="Smart Attendance", layout="wide")

st.title("🎓 Smart Attendance System")

run = st.checkbox("Start Camera")
```

```
col1, col2 = st.columns(2)

frame_placeholder = col1.empty()
status_placeholder = col2.empty()


# Stability buffer
recent_predictions = []


def get_stable_name(new_name):
    recent_predictions.append(new_name)
    if len(recent_predictions) > 5:
        recent_predictions.pop(0)
    return Counter(recent_predictions).most_common(1)[0][0]


# Prevent duplicate marking
marked = set()


# CAMERA LOOP
if run:
    cap = cv2.VideoCapture(0)
    st.success("Camera started")


    while run:
        ret, frame = cap.read()
        if not ret:
            break

        results = yolo(frame)[0]
        detected_name = "No Face"

        for box in results.bboxes:
            cls = int(box.cls[0])
```

```
# Only detect person
if cls != 0:
    continue

x1, y1, x2, y2 = map(int, box.xyxy[0])

# FIXED cropping
face = frame[y1:y2, x1:x2]

if face.size == 0:
    continue

face_tensor = preprocess_face(face)
emb = get_embedding(face_tensor)

name = match_face(emb, database)
name = get_stable_name(name)

if name != "Unknown" and name not in marked:
    mark(name)
    marked.add(name)

detected_name = name

cv2.rectangle(frame, (x1, y1), (x2, y2), (0,255,0), 2)
cv2.putText(frame, name, (x1, y1-10),
            cv2.FONT_HERSHEY_SIMPLEX, 0.9, (0,255,0), 2)

frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
frame_placeholder.image(frame)
```

```
status_placeholder.markdown(f"### Detected: {detected_name}")
```

```
cap.release()
```

```
# TODAY ATTENDANCE
```

```
st.subheader("Today's Attendance")
```

```
query_today = """
```

```
SELECT * FROM attendance
```

```
WHERE date = CURDATE()
```

```
"""
```

```
df_today = pd.read_sql(query_today, conn)
```

```
st.dataframe(df_today)
```

```
# PERSON-WISE COUNT
```

```
st.subheader(" Person-wise Attendance")
```

```
query_count = """
```

```
SELECT name, COUNT(*) as total_present
```

```
FROM attendance
```

```
GROUP BY name
```

```
"""
```

```
df_count = pd.read_sql(query_count, conn)
```

```
st.dataframe(df_count)
```

```
st.bar_chart(df_count.set_index("name"))
```

```
# MONTHLY ANALYTICS
```

```
st.subheader("Monthly Analytics")
```

```
query_month = ""
```

```
SELECT name, COUNT(*) as present_days
```

```
FROM attendance
```

```
WHERE MONTH(date) = MONTH(CURDATE())
```

```
GROUP BY name
```

```
""
```

```
df_month = pd.read_sql(query_month, conn)
```

```
TOTAL_DAYS = 5
```

```
df_month["absent_days"] = TOTAL_DAYS - df_month["present_days"]
```

```
df_month["attendance_%"] = (df_month["present_days"] / TOTAL_DAYS) * 100
```

```
st.dataframe(df_month)
```

```
st.bar_chart(df_month.set_index("name")[["present_days", "absent_days"]])
```

6. Features

- Automated attendance marking
- Real-time face recognition
- Duplicate prevention using tracking
- Daily and monthly attendance analytics
- Visual dashboards (charts & tables)
- Scalable for Smart Classroom environments

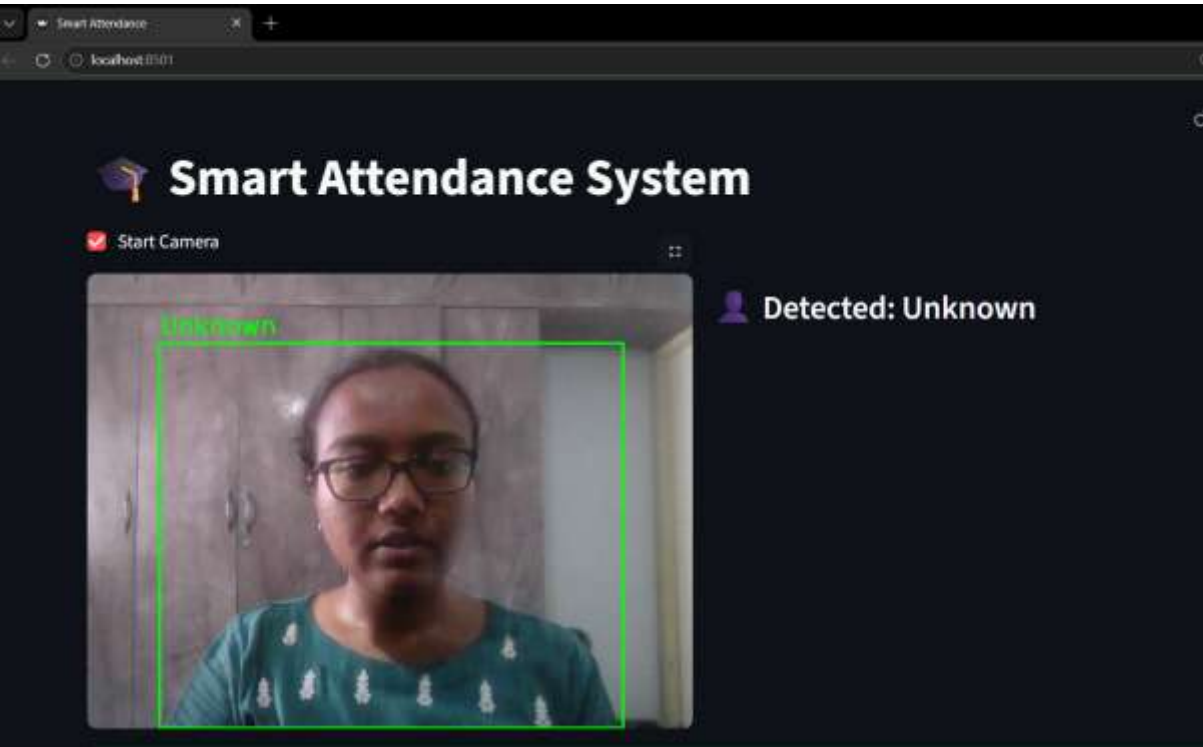
7. Results and Analysis

The system successfully:

- Detects and recognizes multiple faces in real-time
- Reduces manual workload significantly

- Provides accurate attendance reports
- Enhances transparency and efficiency

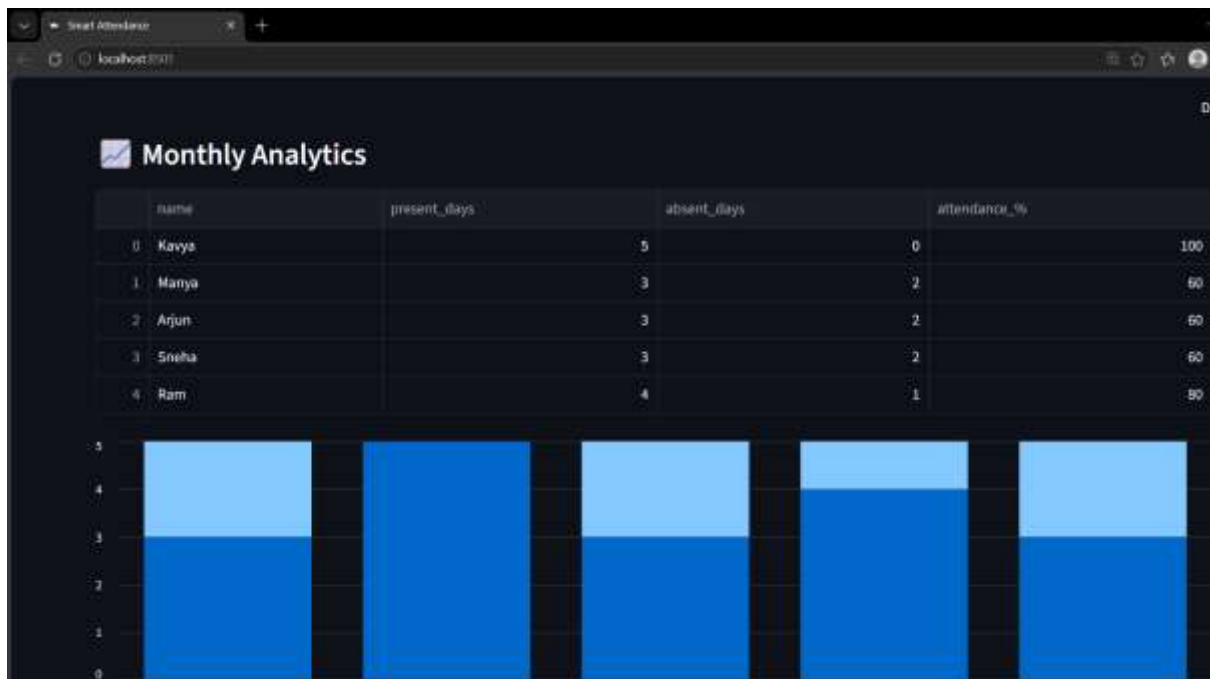
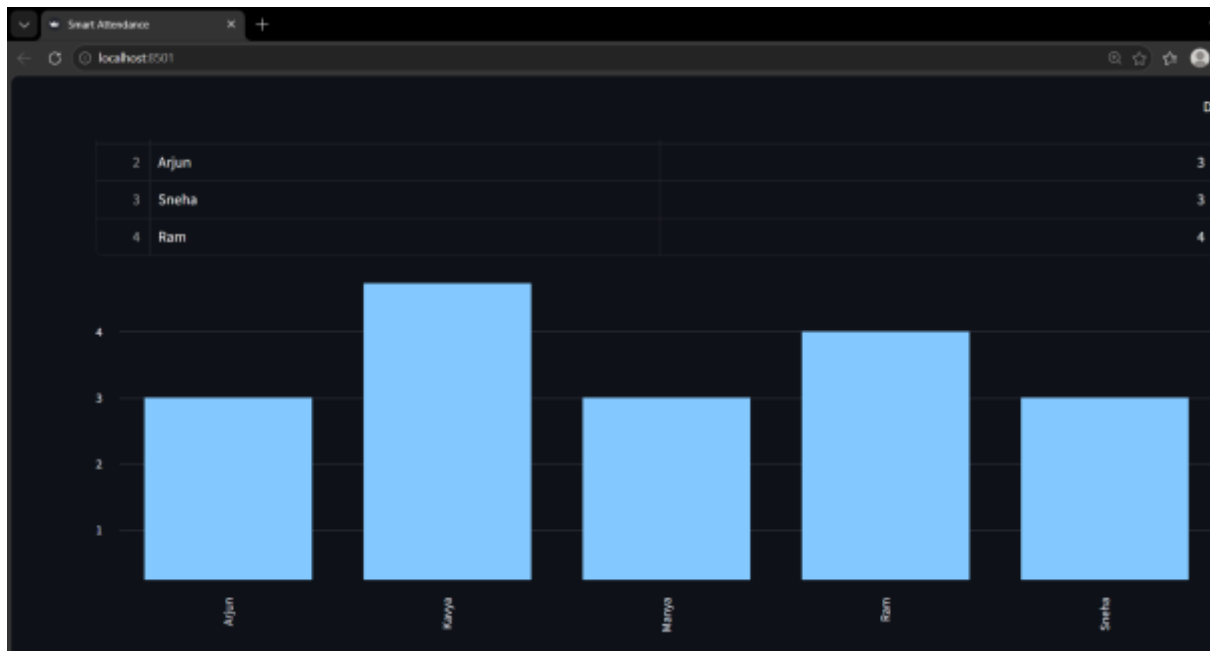
Output:



The screenshot shows the 'Smart Attendance System' web application displaying attendance reports. The interface has a dark theme. At the top, there's a title 'Today's Attendance' with a calendar icon. Below the title, there's a table with columns 'id', 'name', 'date', and 'time'. The table contains two rows of data. Below the table, there's a section titled 'Person-wise Attendance' with a person icon. Below this section, there's a table with columns 'name' and 'total_present'. The table contains five rows of data.

id	name	date	time
0	20 Ram	2026-04-18	14 hours
1	21 Kavya	2026-04-18	14 hours

name	total_present
0 Kavya	5
1 Manya	3
2 Arjun	3
3 Sneha	3
4 Ram	4





MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

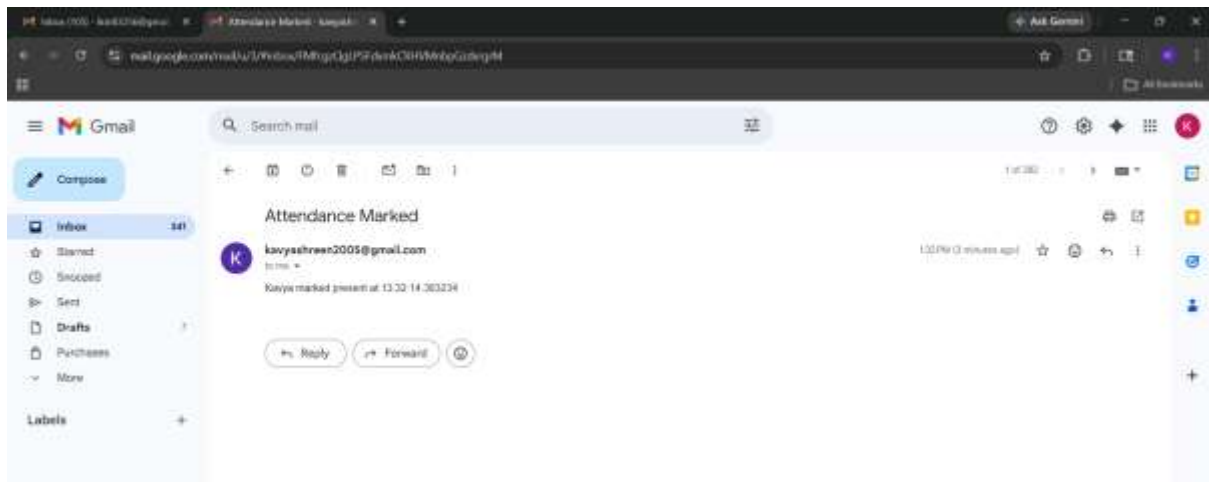
- attendance_db
 - Tables
 - attendance
 - Views
 - Stored Procedures
 - Functions
- companydb
- customerdb
- datasets
- departments
 - Tables
 - Views
 - Stored Procedures
 - Functions
- productsdb
- sys

SQL File 3: SQL File 2

```
1 SELECT * FROM attendance
2 WHERE date = CURDATE();
```

Result Grid

id	name	date	time
20	Ram	2025-04-18	13:31:55
21	Kavya	2025-04-18	13:32:14



8. Conclusion

The proposed AI-powered Smart Attendance System demonstrates how computer vision and deep learning can transform traditional classroom processes. By integrating face recognition with real-time analytics, the system contributes to the evolution of Smart Classrooms and Digital Learning Platforms. It enhances efficiency, accuracy, and scalability, making it a vital component of next-generation Educational ERP systems.